

Scoring dimensions: definitions and data sources

Dimension	Weight	Description & data sources
Nuclear localization	x4	UniProt subcellular location, HPA IF images, GO-CC (IDA/IMP evidence). Nucleus + chromatin specificity.
Protein size	x1	Amino acid length from UniProt. Optimal range 300-800 aa scored highest (experimental tractability).
Research novelty	x5	PubMed publication count. < 50 papers = high novelty. Prioritizes under-studied candidates.
3D structure	x3	AlphaFold pLDDT confidence, PDB entries, domain-level structural quality assessment.
Regulatory domains	x2	InterPro / SMART / Pfam annotation. Chromatin-binding, DNA-binding, and enzymatic domains scored highest.
PPI network	x3	IntAct / STRING / BioGRID interaction data. Weighted by regulatory partner enrichment in chromatin/RNA.
Cross-validation	bonus	0 to +3 concordance bonus for multi-source agreement across all databases (UniProt, HPA, GO, PDB).